

Quantum computing

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1 Mathematical definitions of physical concepts

Definition 1.0.1. A **state space** is a complex Hilbert space, a unit vector in $\mathbb{H}$ is a **state**. A **qubit** is the state space $\mathbb{H} \cong (\mathbb{C}^2)^{\otimes n}$ for some $n > 0$.¹ Sometimes to specify that a qubit is $\mathbb{C}^2$ we will say the qubit is a **single qubit**.

The trace of an operator is part of the mathematical formalisation of a *measurement* (Definition 1.0.4), how the trace becomes part of the formalism hinges crucially on Lemma 1.0.3 below.

Throughout, $\mathbb{H}$ is a fixed state space.

Definition 1.0.2. The **trace** of an operator $T : \mathbb{H} \longrightarrow \mathbb{H}$ is the trace of any (and hence all) matrix representation of T .

The trace of an operator can be written using bra-ket notation: let $|v_1\rangle, \dots, |v_n\rangle$ be an orthogonal basis for $\mathbb{H}$,² and let A be an operator on $\mathbb{H}$ and write $A|v_j\rangle = a_{1j}|v_1\rangle + \dots + a_{nj}|v_n\rangle$, then we have

$$\langle v_i | A | v_j \rangle = a_{ij} \tag{1}$$

¹Standard usage: $\mathbb{C}^2$ is a single-qubit state space, and $(\mathbb{C}^2)^{\otimes n}$ is an n -qubit state space. See Section 5.

²The formula $\text{Tr}(A) = \sum_i \langle v_i | A | v_i \rangle$ requires the basis to be *orthonormal*, not merely orthogonal. See Section 5.

and so

$$\text{Trace } A = \sum_{i=1}^n \langle v_i | A | v_i \rangle \quad (2)$$

The trace of A precomposed with a projector has a particularly nice form: let $|\psi\rangle$ be a unit vector in $\mathbb{H}$ and let $\{|\psi\rangle, |v_2\rangle, \dots, |v_n\rangle\}$ be an orthogonal basis for $\mathbb{H}$. Then

$$\begin{aligned} \text{Trace}(A |\psi\rangle \langle \psi|) &= \langle \psi | A |\psi\rangle \langle \psi | \psi \rangle + \sum_{i=2}^n \langle v_i | A |\psi\rangle \langle \psi | v_i \rangle \\ &= \langle \psi | A |\psi\rangle \end{aligned}$$

In a statement:

Lemma 1.0.3. *Let A be an operator on a Hilbert space $\mathbb{H}$ and let $|\psi\rangle \in \mathbb{H}$ be a unit vector in $\mathbb{H}$. We have the following formula:*

$$\text{Trace}(A |\psi\rangle \langle \psi|) = \langle \psi | A |\psi\rangle \quad (3)$$

Proof. Use the Gram-Schmidt process to extend $\{|\psi\rangle\}$ to an orthonormal basis, then apply the previous discussion. $\square$

Definition 1.0.4. ³ A **measurement** on $\mathbb{H}$ is a finite family of linear operators $\{M_m : \mathbb{H} \longrightarrow \mathbb{H}\}_{m \in \mathcal{M}}$ satisfying

$$\sum_{m \in \mathcal{M}} M_m^\dagger M_m = I \quad (4)$$

Associated to every measurement and state vector $|\psi\rangle$ there is a value

$$p(m) := \langle \psi | M_m^\dagger M_m | \psi \rangle \in [0, 1] \quad (5)$$

which is understood as the probability $p(m)$ of outcome m on measurement M_m . Notice by Lemma 1.0.3 that

$$p(m) = \text{Trace}(M_m^\dagger M_m |\psi\rangle \langle \psi|) \quad (6)$$

Also, there is a **resulting state**, which is the state

$$\frac{M_m |\psi\rangle}{\sqrt{p(m)}} \quad (7)$$

understood as the state of $\mathbb{H}$ after measuring it with $\{M_m\}_{m \in \mathcal{M}}$.

Definition 1.0.5. ⁴ A linear transformation P is a **projector** if $P^2 = P$.

Recall that Hermitian operators M are in particular *Normal*, ie, satisfy $MM^\dagger = M^\dagger M$, and so are diagonalisable (see [2]).

Definition 1.0.6. ⁵ A **projective measurement** on $\mathbb{H}$ is a Hermitian operator $M : \mathbb{H} \longrightarrow \mathbb{H}$ with spectral decomposition

$$M = \sum_{m \in \mathcal{M}} m P_m \quad (8)$$

³The completeness sum should read $\sum_{m \in \mathcal{M}} M_m^\dagger M_m = I$ (subscripts on each M_m). The post-measurement state $M_m |\psi\rangle / \sqrt{p(m)}$ is defined only when $p(m) > 0$. See Section 5.

⁴In quantum mechanics “projector” means *orthogonal* projector: $P^2 = P$ and $P^\dagger = P$. The bare condition $P^2 = P$ defines an algebraic idempotent. See Section 5.

⁵The body has “ $m \setminus \text{in } \mathcal{M}$ ”; should read $m \in \mathcal{M}$. Post-measurement state requires $p(m) > 0$. See Section 5.

where P_m is the projector onto the m -eigenspace. Associated to value m in $\mathcal{M}$ and state $|\psi\rangle$ there is a value

$$p(m) := \langle \psi | P_m | \psi \rangle \quad (9)$$

which is understood as the probability $p(m)$ of outcome m on measurement P_m . Also, there is a **resulting state**:

$$\frac{P_m |\psi\rangle}{\sqrt{p(m)}} \quad (10)$$

understood as the state of $|\psi\rangle$ after measuring it with P_m .

Lemma 1.0.7. ⁶ *Let $W \subseteq V$ be a subspace of a Hilbert space V , and let $U : W \rightarrow V$ be a unitary operator. Then U extends to a unitary U' operator on all of V .*

Proof sketch. Define $U' = \text{Proj}_W \circ U$. □

The relationship between projective measurements and measurements is the following:

Proposition 1.0.8. ⁷ *Let $\{M_m\}_{m \in \mathcal{M}}$ be a measurement on $\mathbb{H}$. Then there exists a projective measurement M , a state space Q , and a unitary operator $U : Q \otimes \mathbb{H} \rightarrow Q \otimes \mathbb{H}$ such that the measurement $\{M_m\}_{m \in \mathcal{M}}$ is modelled by M and U (the meaning of modelled is provided by the proof).*

Proof. Let Q be the free Hilbert space of dimension $|\mathcal{M}|$, the cardinality of $\mathcal{M}$. Let $|\varphi\rangle$ be an arbitrary, fixed vector inside Q , and define

$$U : \mathbb{H} \otimes \text{Span} |\varphi\rangle \rightarrow \mathbb{H} \otimes Q$$

given by

$$U |\psi\rangle \otimes |\varphi\rangle = \sum_{m \in \mathcal{M}} M_m |\psi\rangle \otimes |m\rangle \quad (11)$$

We first prove this is unitary. We calculate:

$$\begin{aligned} \langle \psi_1 | \otimes \langle \varphi | U^\dagger U | \psi_2 \rangle \otimes | \psi_1 \rangle &= (U | \psi_1 \rangle \otimes | \varphi \rangle)^\dagger U | \psi_2 \rangle \otimes | \varphi \rangle \\ &= \left(\sum_{m \in \mathcal{M}} M_m | \psi_1 \rangle \otimes | m \rangle \right)^\dagger \sum_{m' \in \mathcal{M}} M_{m'} | \psi_2 \rangle \otimes | m' \rangle \\ &= \sum_{m \in \mathcal{M}} \langle \psi_1 | M_m^\dagger \langle m | \sum_{m' \in \mathcal{M}} M_{m'} | \psi_2 \rangle \otimes | m' \rangle \\ &= \sum_{m \in \mathcal{M}} \sum_{m' \in \mathcal{M}} \langle \psi_1 | M_m^\dagger M_{m'} | \psi_2 \rangle \langle m | m' \rangle \\ &= \sum_{m \in \mathcal{M}} \langle \psi_1 | M_m^\dagger M | \psi_2 \rangle \\ &= \langle \psi_1 | \psi_2 \rangle \end{aligned}$$

Thus U extends to a unitary operator $U : \mathbb{H} \otimes Q \rightarrow \mathbb{H} \otimes Q$.

Now consider the following projective measurement on $\mathbb{H} \otimes Q$:

$$P_m := I_q \otimes |m\rangle \langle m| \quad (12)$$

⁶This statement is incorrect: a unitary $U : W \rightarrow V$ forces $W = V$ (in finite dimensions), so the conclusion is trivial. The intended hypothesis is that U is an *isometry*. The proof $U' = \text{Proj}_W \circ U$ is also wrong (domain/codomain mismatch). See Section 5 for a correct statement and proof.

⁷The proof switches between $Q \otimes \mathbb{H}$ and $\mathbb{H} \otimes Q$, omits subscripts on $M_m^\dagger M_{m'}$, writes $I_q \otimes |m\rangle \langle m|$ instead of $I_{\mathbb{H}} \otimes |m\rangle \langle m|$, and has a wrong final ket in the inner-product calculation. See Section 5 for a clean version.

Then the probability outcome n occurs is:

$$\begin{aligned}
p(n) &= \langle \psi | \otimes \langle \varphi | U^\dagger P_n U | \psi \rangle \otimes | \varphi \rangle \\
&= (U | \psi \rangle \otimes | \varphi \rangle)^\dagger P_n U | \psi \rangle \otimes | \varphi \rangle \\
&= \left(\sum_{m \in \mathcal{M}} M_m | \psi \rangle \otimes | m \rangle \right)^\dagger P_n \sum_{m' \in \mathcal{M}} M_{m'} | \psi \rangle \otimes | m \rangle \\
&= \left(\sum_{m \in \mathcal{M}} \langle \psi | M_m^\dagger \langle m | \right) I_Q \otimes | n \rangle \langle n | \sum_{m' \in \mathcal{M}} M_{m'} | \psi \rangle \otimes | m \rangle \\
&= \left(\sum_{m \in \mathcal{M}} \langle \psi | M_m^\dagger \langle m | \right) \sum_{m' \in \mathcal{M}} M_{m'} | \psi \rangle \otimes | n \rangle \langle n | m \rangle \\
&= \sum_{m \in \mathcal{M}} \langle \psi | M_m^\dagger \langle m | M_m | \psi \rangle \otimes | n \rangle \\
&= \sum_{m \in \mathcal{M}} \langle \psi | M_m^\dagger M_n | \psi \rangle \langle m | n \rangle \\
&= \langle \psi | M_n^\dagger M_n | \psi \rangle
\end{aligned}$$

□

Now we define the *evolution* of a state space:

Definition 1.0.9. Let $\mathbb{H}$ be a state space. A **single step time evolution** of $\mathbb{H}$ is a unitary operator U on $\mathbb{H}$. A **single step time evolution** of a state vector $|\psi\rangle$ with respect to U is the pair $(|\psi\rangle, U|\psi\rangle)$.

An **evolution** of $\mathbb{H}$ is a sequence of unitary operators $(U_1, \dots, U_n)$ on $\mathbb{H}$, an **evolution** of a state vector $|\psi\rangle$ with respect to the evolution $(U_1, \dots, U_n)$ is the sequence $(|\psi\rangle, U_1|\psi\rangle, \dots, U_n \dots U_1|\psi\rangle)$.

Lastly, the *composite* of two state spaces:

Definition 1.0.10. Let $\mathbb{H}_1, \mathbb{H}_2$ be two state spaces. The **composite state space** is $\mathbb{H}_1 \otimes \mathbb{H}_2$. We often describe a state of $\mathbb{H}_1 \otimes \mathbb{H}_2$ using the terminology “ $\mathbb{H}_1$ is in state $|\psi_1\rangle$ and $\mathbb{H}_2$ is in state $|\psi_2\rangle$ ”, this simply describes the state $|\psi_1\rangle \otimes |\psi_2\rangle \in \mathbb{H}_1 \otimes \mathbb{H}_2$.

2 The density operator

Definition 2.0.1. ⁸ Let $\mathbb{H}$ be a state space. An **ensemble of states** is a finite set $\{(p_1, |\psi_1\rangle), \dots, (p_n, |\psi_n\rangle)\}$ of tuples consisting of state vectors $|\psi_1\rangle, \dots, |\psi_n\rangle$ with associated probabilities $p_1, \dots, p_n$, that is, $p_i \in [0, 1] \subseteq \mathbb{R}$. We require $\sum_{i=1}^n p_i = 1$.

If $n = 1$ then the ensemble is a **pure state**, otherwise it is a **mixed state**.

Associated to an ensemble of states is the **density operator** (sometimes **density matrix**, sometimes **state**)

$$\sum_{i=1}^n p_i |\psi_i\rangle \langle \psi_i| \tag{13}$$

We see now that Section 1 concerned itself with pure states, where we identify a state vector $|\psi\rangle$ with the operator $|\psi\rangle \langle \psi|$ (that is, we identify the vector $|\psi\rangle$ with the projection onto this vector). We now describe how to generalise the Definitions of Section 1 to the case of mixed states.

⁸Purity is a property of the density operator ρ , not of the ensemble. ρ is pure iff $\rho = |\psi\rangle \langle \psi|$, equivalently $\text{Tr}(\rho^2) = 1$. An ensemble of length > 1 may still produce a pure ρ . See Section 5.

The Definition of a measurement remains identical, notice however that to a measurement $\{M_m : \mathbb{H} \longrightarrow \mathbb{H}\}_{m \in \mathcal{M}}$ the associated value $p(m)$ is now a sum (NB: this is a *definition* which is motivated by the law of total probability):

$$\begin{aligned} p(m) &= \sum_{i=1}^n p_i p(m | i) \\ &= \sum_{i=1}^n p_i \langle \psi_i | M_m^\dagger M_m | \psi_i \rangle \\ &= \sum_{i=1}^n p_i \text{Trace}(M_m^\dagger M_m | \psi_i \rangle \langle \psi_i |) \\ &= \text{Trace}(M_m^\dagger M_m \rho) \end{aligned}$$

where the last equality follows from linearity of the Trace.

The resulting density operator is then *defined* to be

$$\rho_m := \sum_{i=1}^n p(i | m) \frac{M | \psi_i \rangle \langle \psi_i | M^\dagger}{p(m | i)} \quad (14)$$

we then use Bayes Theorem:

$$p(i | m) / p(m | i) = p_i / p(m) \quad (15)$$

to obtain:

$$\rho_m = \sum_{i=1}^n p_i \frac{M_m | \psi_i \rangle \langle \psi_i | M_m^\dagger}{p(m)} = \sum_{i=1}^n \frac{M_m \rho M_m^\dagger}{\text{Trace}(M_m^\dagger M_m \rho)} \quad (16)$$

We arrive at:

Definition 2.0.2. ⁹ A **measurement** on $\mathbb{H}$ is a family of linear operators $\{M_m : \mathbb{H} \longrightarrow \mathbb{H}\}_{m \in \mathcal{M}}$ satisfying

$$\sum_{m \in \mathcal{M}} M_m^\dagger M_m = I \quad (17)$$

Associated to every measurement and density operator ρ there is a value

$$p(m) = \text{Trace}(M_m^\dagger M_m \rho) \quad (18)$$

which is understood as the probability $p(m)$ of outcome m on measurement $\{M_m\}_{m \in \mathcal{M}}$.

Also, there is a **resulting density operator**:

$$\rho_m := \frac{M_m^\dagger \rho M_m}{\text{Trace}(M_m^\dagger M_m \rho)} \quad (19)$$

Lemma 2.0.3. For a pure state density operator $|\psi\rangle \langle \psi|$ Definitions 2.0.2 and 1.0.4 agree once $|\psi\rangle \langle \psi|$ has been identified with $|\psi\rangle$.

Now, the Definition of time evolution:

Definition 2.0.4. Let $\mathbb{H}$ be a state space. A **single step time evolution** on $\mathbb{H}$ is a unitary operator $U : \mathbb{H} \longrightarrow \mathbb{H}$. A **single step time evolution** of a density operator ρ with respect to U is the pair $(\rho, U\rho U^\dagger)$.

An **evolution** of $\mathbb{H}$ is a sequence of unitary operators $(U_1, \dots, U_n)$ on $\mathbb{H}$, an **evolution** of a density operator ρ with respect to the evolution $(U_1, \dots, U_n)$ is the sequence $(\rho, U\rho U^\dagger, \dots, U_n \dots U_1 \rho U_1^\dagger \dots U_n^\dagger)$.¹⁰

⁹The post-measurement density operator has the wrong operator order: it should be $\rho_m = M_m \rho M_m^\dagger / \text{Tr}(M_m^\dagger M_m \rho)$, not $M_m^\dagger \rho M_m / \text{Tr}(M_m^\dagger M_m \rho)$. With the correct ordering it agrees with the pure-state rule. The completeness condition is also missing m -subscripts. See Section 5.

¹⁰The first evolved state should be $U_1 \rho U_1^\dagger$, not $U \rho U^\dagger$. See Section 5.

Finally:

Definition 2.0.5. Let $\mathbb{H}_1, \mathbb{H}_2$ be two state spaces. The **composite state space** is $\mathbb{H}_1 \otimes \mathbb{H}_2$. We often describe a state of $\mathbb{H}_1 \otimes \mathbb{H}_2$ using the terminology “ $\mathbb{H}_1$ is in state ρ_1 and $\mathbb{H}_2$ is in state ρ_2 ”, this simply describes the state $\rho_1 \otimes \rho_2 \in \mathbb{H}_1 \otimes \mathbb{H}_2$.

Let ρ be an arbitrary, positive operator on a finite dimensional state space $\mathbb{H}$. Then since ρ is positive, it is Hermitian, and thus normal. We thus obtain a spectral decomposition:

$$\rho = \sum_{i=1}^n \lambda_i |\psi_i\rangle \langle \psi_i| \quad (20)$$

Thus, if $\text{Trace } \rho = 1$, we have a well defined density operator with respect to the ensemble $\{\lambda_i, |\psi_i\rangle\}$. That every density operator is positive and has trace equal to one is an easy exercise, we arrive at:

Definition 2.0.6 (Intrinsic definition of density operator). Let $\mathbb{H}$ be finite dimensional. A **density operator** is a positive operator $\rho : \mathbb{H} \rightarrow \mathbb{H}$ with trace equal to 1.

Note that we have not described a canonical form of an ensemble, and so it is reasonable to expect different ensembles to give rise to the same density operator, indeed this is so (see [?, Page 103] for an explicit example). This leads to a natural question: when do two different ensembles give rise to the same density operator? This is answered by Proposition 2.0.8, but first we make an observation, which may be skipped, to make the normalisation in the statement of Proposition 2.0.8 more transparent:

Observation 2.0.7. Let $\mathbb{H}$ be of dimension n . Let $\sum_{i=1}^m p_i |\psi_i\rangle \langle \psi_i|$ be a density operator associated to an ensemble. Since this is diagonalisable, we write

$$\sum_{i=1}^m p_i |\psi_i\rangle \langle \psi_i| = \sum_{j=1}^n \lambda_j |\varphi_j\rangle \langle \varphi_j| \quad (21)$$

where $|\varphi_1\rangle, \dots, |\varphi_n\rangle$ is an orthonormal basis for $\mathbb{H}$. Write for each i :

$$|\psi_i\rangle = \sum_{j=1}^n \alpha_{ij} |\varphi_j\rangle \quad (22)$$

Substituting (22) into (21) to obtain:

$$\begin{aligned} \sum_{j=1}^n \lambda_j |\varphi_j\rangle \langle \varphi_j| &= \sum_{i=1}^m p_i \sum_{j,k=1}^n \alpha_{ij} \overline{\alpha_{ki}} |\varphi_j\rangle \langle \varphi_k| \\ &= \sum_{j,k=1}^n \left(\sum_{i=1}^m p_i \alpha_{ij} \overline{\alpha_{ki}} \right) |\varphi_j\rangle \langle \varphi_k| \end{aligned}$$

which implies

$$\lambda_j = \delta_{jk} \sum_{i=1}^m p_i \alpha_{ij} \alpha_{ij}^\dagger \quad (23)$$

We mould this calculation to give the result of Proposition 2.0.8 by normalisation, indeed we replace (21) with

$$\sqrt{p_i} |\psi_i\rangle = \sum_{j=1}^n \alpha_{ij} \sqrt{\lambda_j} |\varphi_j\rangle \quad (24)$$

and show that α_{ij} is unitary.

Proposition 2.0.8. ¹¹ Let $\{p_i |i\rangle \langle i|\}, \{q_j |j\rangle \langle j|\}$ be two ensembles and denote by ρ_i, ρ_j their respective induced density operators. We Then $\rho_i = \rho_j$ if and only if for each $i = 1, \dots, m$ there exists a unitary matrix u_{jk}^i such that

$$\sqrt{p_i} |\psi_i\rangle = \sum_{j=1}^n \alpha_{ij} \sqrt{q_j} |\varphi_j\rangle$$

with α_{ij} a unitary matrix.

Proof. Write $\sum_{j=1}^n \lambda_j |\varphi_j\rangle$ for the diagonalisation of ρ_i and write

$$\sqrt{p_i} |\psi_i\rangle = \sum_{j=1}^n \sqrt{\lambda_j} |\varphi_j\rangle \quad (25)$$

Then

$$\begin{aligned} \sum_{i=1}^m p_i |\psi_i\rangle \langle \psi_i| &= \sum_{i=1}^m \sqrt{p_i} |\psi_i\rangle \sqrt{p_i} \langle \psi_i| \\ &= \sum_{i=1}^m \sum_{j,k=1}^n \alpha_{ij} \overline{\alpha_{kj}} \sqrt{\lambda_j} |\varphi_j\rangle \sqrt{\lambda_k} \langle \varphi_k| \\ &= \sum_{j,k=1}^n \left(\alpha_{ij} \overline{\alpha_{kj}} \right) \sqrt{\lambda_j} |\varphi_j\rangle \sqrt{\lambda_k} \langle \varphi_k| \end{aligned}$$

We have by assumption definition that $\sum_{i=1}^m p_i |\psi_i\rangle \langle \psi_i| = \sum_{i=1}^n \lambda_i |\varphi_i\rangle \langle \varphi_i| = \sum_{i=1}^n \sqrt{\lambda_i} |\varphi_i\rangle \sqrt{\lambda_i} \langle \varphi_i|$. It follows that α_{ij} is unitary. Applying the same logic to ρ_j we see that both ρ_i, ρ_j are related to $\sum_{j=1}^n \lambda_j |\varphi_j\rangle \langle \varphi_j|$ and so we are done. $\square$

2.1 Partial trace

For an introduction to the partial trace operator, see [3]

Example 2.1.1. We calculate the partial trace of the operator

$$\rho := \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right) \left(\frac{\langle 00| + \langle 11|}{\sqrt{2}} \right) = \frac{|00\rangle \langle 00| + |00\rangle \langle 11| + |11\rangle \langle 00| + |11\rangle \langle 11|}{2} \quad (26)$$

We denote the first copy of $\mathbb{C}^2$ simply by 1 and the second copy of $\mathbb{C}^2$ by 2.

First consider $\text{Trace}_2(|00\rangle \langle 00|)$. We have, where we write $E_{ij} : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ for the linear map which maps the i^{th} basis vector to the j^{th} basis vector, and similarly for F_{ij} (just applied to the second copy of $\mathbb{C}^2$)

$$|00\rangle \langle 00| = E_{00} \otimes F_{00} \quad (27)$$

and so

$$\text{Trace}_2(|00\rangle \langle 00|) = \text{Trace}(F_{00}) E_{00} = |0\rangle \langle 0| \quad (28)$$

Similarly,

$$|11\rangle \langle 00| = E_{01} \otimes F_{01}, \quad |00\rangle \langle 11| = E_{10} \otimes F_{10}, \quad |11\rangle \langle 11| = E_{11} \otimes F_{11} \quad (29)$$

¹¹This statement is mathematically incoherent: the notation $\{p_i |i\rangle \langle i|\}$ is not standard ensemble notation; indices, kets/bras, and quantifiers are conflated; and the proof omits bras in spectral decompositions. See Section 5 for the standard unitary-freedom theorem for ensembles.

and so

$$\text{Trace}_2(|11\rangle\langle 00|) = \text{Trace}(F_{00})E_{11} = 0 \quad (30)$$

$$\text{Trace}_2(|11\rangle\langle 00|) = \text{Trace}(F_{00})E_{11} = 0 \quad (31)$$

$$\text{Trace}_2(|11\rangle\langle 11|) = \text{Trace}(F_{11})E_{11} = |1\rangle\langle 1| \quad (32)$$

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we thus have

$$\text{Trace}_2(\rho) = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2} = I/2 \quad (33)$$

We present another way defining the partial trace:

Observation 2.1.2. ¹³ Let V be a finite dimensional vector space. The functor $V \otimes _$ admits a right adjoint which is given by the functor $_ \otimes V^*$:

$$\text{Hom}(V \otimes U, W) \cong \text{Hom}(V, W \otimes V^*) \quad (34)$$

2.2 Quantum operations

Definition 2.2.1. An **open quantum system** is the tensor product of two state spaces $\mathbb{H}_p \otimes \mathbb{H}_e$ where $\mathbb{H}_p$ is the **principal system** and $\mathbb{H}_e$ is the **environment**.

Definition 2.2.2. The **time evolution of an open quantum system** is that of a Definition 1.0.9 where an open quantum system is thought of as a composite system (Definition 1.0.10).

Given an open quantum system $\mathbb{H}_p \otimes \mathbb{H}_e$, and a state $\rho \otimes \rho_e$ a **quantum operation** is an operator $\mathcal{E} : \mathbb{H}_p \longrightarrow \mathbb{H}_p$ ¹⁴ given by the following formula, where U is a unitary operator U on the entire open quantum system $\mathbb{H}_p \otimes \mathbb{H}_e$:

$$\mathcal{E}(\rho) = \text{Trace}_{\mathbb{H}_e}(U(\rho \otimes \rho_e)U^\dagger) \quad (35)$$

Remark 2.2.3. Let $\mathbb{H}_p \otimes \mathbb{H}_e$ be an open quantum system and let $|1\rangle, \dots, |n\rangle$ be a basis for $\mathbb{H}_e$. We think of $|1\rangle, \dots, |n\rangle$ as operators $\mathbb{C} \longrightarrow V$ and write $\text{id} \otimes |i\rangle$ for the composite $\mathbb{H}_p \longrightarrow \mathbb{H}_p \otimes \mathbb{C} \longrightarrow \mathbb{H}_p \otimes \mathbb{H}_e$. Also, we denote the multiplication map $\mathbb{H}_p \otimes k \longrightarrow \mathbb{H}_p$ defined by $w \otimes x \longmapsto xw$ by Mult . We have for any operator $f : \mathbb{H}_p \otimes \mathbb{H}_e \longrightarrow \mathbb{H}_p \otimes \mathbb{H}_e$ that

$$\text{Trace}_{\mathbb{H}_e} f = \text{Mult} \left(\sum_{i=1}^n (\text{id} \otimes \langle i|) f (\text{id} \otimes |i\rangle) \right) \quad (36)$$

See [3] for more details.

We let $|1\rangle, \dots, |n\rangle$ be a basis for $\mathbb{H}_e$ and use (36) to rewrite (35) in the special case where $\rho_e = |e_0\rangle\langle e_0|$.¹⁵We have

$$\mathcal{E}(\rho) = \text{Trace}_{\mathbb{H}_e}(U(\rho \otimes |i\rangle\langle i|)U^\dagger) \quad (37)$$

$$= \text{Mult} \left(\sum_{i=1}^n (\text{id} \otimes \langle i|) (U(\rho \otimes |i\rangle\langle i|)U^\dagger) (\text{id} \otimes |i\rangle) \right) \quad (38)$$

¹²The $|11\rangle\langle 00|$ line is duplicated and the trace entries are mislabelled. The correct rows are $\text{Tr}_2(|11\rangle\langle 00|) = \text{Tr}(F_{01})E_{01} = 0$, $\text{Tr}_2(|00\rangle\langle 11|) = \text{Tr}(F_{10})E_{10} = 0$, $\text{Tr}_2(|11\rangle\langle 11|) = \text{Tr}(F_{11})E_{11} = |1\rangle\langle 1|$. See Section 5.

¹³The standard tensor-Hom adjunction is $\text{Hom}(V \otimes U, W) \cong \text{Hom}(U, V^* \otimes W)$. The right adjoint of $V \otimes _$ is $V^* \otimes _$. See Section 5.

¹⁴Type-incorrect: a quantum operation is a linear map on *operators*, $\mathcal{E} : L(\mathbb{H}_p) \rightarrow L(\mathbb{H}_p)$, not on the Hilbert space itself. Trace-preserving CP such maps are called quantum channels. See Section 5.

¹⁵The derivation conflates the summation index i with the fixed environment state, writing $\rho \otimes |i\rangle\langle i|$ where $\rho \otimes |e_0\rangle\langle e_0|$ is intended. See Section 5.

Now we make the observation that

$$\rho \otimes |i\rangle \langle i| = (\rho \otimes \text{id})(\text{id} \otimes |e_0\rangle \langle e_0|)(\text{id} \otimes \langle e_0|) \quad (39)$$

$$= (\text{id} \otimes |e_0\rangle \langle e_0|)(\rho \otimes \text{id})(\text{id} \otimes \langle e_0|) \quad (40)$$

Substituting (40) into (38) we obtain:

$$\begin{aligned} & \text{Mult} \left(\sum_{i=1}^n (\text{id} \otimes \langle i|)(U(\rho \otimes |i\rangle \langle i|)U^\dagger)(\text{id} \otimes |i\rangle) \right) \\ &= \text{Mult} \left(\sum_{i=1}^n (\text{id} \otimes \langle i|)U(\text{id} \otimes |e_0\rangle \langle e_0|)(\rho \otimes \text{id})(\text{id} \otimes \langle e_0|)U^\dagger(\text{id} \otimes |i\rangle) \right) \\ &= \sum_{i=1}^n \text{Mult} \left((\text{id} \otimes \langle i|)U(\text{id} \otimes |e_0\rangle \langle e_0|) \right) \rho \text{Mult} \left((\text{id} \otimes \langle e_0|)U^\dagger(\text{id} \otimes |i\rangle) \right) \\ &= \sum_{i=1}^n E_i \rho E_i^\dagger \end{aligned}$$

where $E_i = \text{Mult} \left((\text{id} \otimes \langle i|)(U((\text{id} \otimes |e_0\rangle \langle e_0|))) \right)$.

We thus have a second Definition of a quantum operation:

Lemma 2.2.4. ¹⁶ *Given an open quantum system $\mathbb{H}_p \otimes \mathbb{H}_e$, a quantum operation $\mathcal{E}$ is a finite set of operators $\{E_1, \dots, E_n\}$ where each E_i is an operator on the principal system $\mathbb{H}_p$ and the collection $\mathcal{E}$ satisfies the **completeness condition**, that*

$$\sum_{i=1}^n E_i^\dagger E_i = I \quad (41)$$

Notice that Definition 2.2.4 does not include the data of U . Indeed, given just the principle system and the operator elements $\{E_1, \dots, E_n\}$, an environment $\mathbb{H}_e$ and a unitary operator $U : \mathbb{H}_p \otimes \mathbb{H}_e \longrightarrow \mathbb{H}_p \otimes \mathbb{H}_e$ can be defined which give rise to $\{E_1, \dots, E_n\}$ following the calculation of Remark 2.2.3. For details see [1, §8.2.3 page 365].

3 Quantum error correction codes

Throughout this Section, we denote by $\mathbb{H}$ a single qubit $\mathbb{C}^2$.

Definition 3.0.1. ¹⁷ Let $|0\rangle, |1\rangle$ respectively denote the standard basis vectors $(0, 1), (1, 0)$ of the $\mathbb{C}$ -vector space $\mathbb{C}^2$. If $a|0\rangle + b|1\rangle$ is a state and both a and b are non-zero then the state is a **superposition**.

Definition 3.0.2. A **quantum gate** is a unitary matrix in $M_2(\mathbb{C})$, and is thought of as an operator $\mathbb{H} \longrightarrow \mathbb{H}$ (written with respect to the standard basis $|0\rangle, |1\rangle$).

The game of quantum computation is to build a computer out of quantum gates. For some basic examples see [1, §4].

Of all the quantum gates, there are a few which are ubiquitous in the theory:

¹⁶A quantum operation is not itself a finite set of operators; it is a linear map admitting an operator-sum representation. The Kraus operators are not unique. See Section 5.

¹⁷Basis convention is reversed: the standard convention (consistent with $Z = \text{diag}(1, -1)$, used throughout the rest of the document) is $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$. See Section 5.

Definition 3.0.3. We denote the following operators:

$$X := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad H := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (42)$$

The matrices X, Y, Z are the **Pauli matrices**, and H is the **Hadamard matrix**.

We make the passing observation that all of X, Y, Z, H square to the identity matrix. The basis vectors

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (43)$$

are the **Bell states**¹⁸ and are denoted $|+\rangle, |-\rangle$ respectively. Notice that as already stated, $H^2 = I$, so $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$.

Definition 3.0.4. The standard basis $|0\rangle, |1\rangle$ of $\mathbb{H}$ induces a basis of $\mathbb{H}^{\otimes n}$, we denote $|0\rangle \otimes \dots \otimes |0\rangle$ by $|0 \dots 0\rangle$, etc.

Notation 3.0.5. Given a collection of Pauli matrices $W_{i_1}, \dots, W_{i_m} \in \{X, Y, Z\}$ where $0 < i_1 < \dots < i_m < n$ we denote by $W_{i_1} \dots W_{i_m}$ the operator on $\mathbb{H}^{\otimes n}$ given by the tensor product with W_{i_j} in the i_j^{th} slot and with the identity operator in all other slots. For example, the operator Z_1 on $\mathbb{H}^{\otimes 3}$ is the operator $Z \otimes I \otimes I$.

Definition 3.0.6. A **message** is a state $|\psi\rangle \in \mathbb{H}^{\otimes n}$, for some n . An **error** is a pair of states $(|\varphi\rangle, |\psi\rangle)$ where $|\varphi\rangle, |\psi\rangle \in \mathbb{H}^{\otimes n}$ for some n , note that an error may be such that $|\varphi\rangle = |\psi\rangle$. The message $|\varphi\rangle$ is the **intended message** and $|\psi\rangle$ is the **received message**.

Definition 3.0.7.¹⁹ An **n -encoding of a single state** (sometimes just an **encoding**) is an injective map $\iota : \mathbb{H} \longrightarrow \mathbb{H}^{\otimes n}$. An **n -encoding of a message** $|m\rangle \in \mathbb{H}^{\otimes k}$ is an n -encoding ι along with a message $|m\rangle \in \mathbb{H}^{\otimes nk}$ for which there exists $|m'\rangle \in \mathbb{H}^{\otimes k}$ satisfying $\iota^{\otimes k}|m'\rangle = |m\rangle$.

Example 3.0.8. Consider the encoding

$$\text{BitFlip} : \mathbb{H} \longrightarrow \mathbb{H}^{\otimes 3} \quad (44)$$

$$|0\rangle \longmapsto |000\rangle \quad (45)$$

$$|1\rangle \longmapsto |111\rangle \quad (46)$$

then an encoding of a message with respect to this encoding might be $|000111000\rangle$, but could not be $|000111001\rangle$. We call Encoding 44 the **bit flip encoding**, for reasons made clear by Algorithm 3.0.13 below. As another example, we consider the **phase flip encoding**:

$$\text{PhaseFlip} : \mathbb{H} \longrightarrow \mathbb{H}^{\otimes 3}$$

$$|0\rangle \longmapsto |+++ \rangle$$

$$|1\rangle \longmapsto |-- - \rangle$$

Definition 3.0.9. Let $\mathbb{H}$ be a state space. A **quantum error correction code (QECC)** for $\mathbb{H}$ is a subspace $C \subseteq \mathbb{H}$.

Example 3.0.10. Continuing with Example 3.0.8, two QECCs are given by

$$\text{im}(\text{BitFlip}), \quad \text{and} \quad \text{im}(\text{PhaseFlip}) \quad (47)$$

¹⁸These are not Bell states. They are the *Hadamard-basis* (or *X-basis*) states $|\pm\rangle$. Bell states are two-qubit entangled states such as $(|00\rangle + |11\rangle)/\sqrt{2}$. See Section 5.

¹⁹A quantum encoding should be a linear isometry, not merely an injective map; otherwise superpositions and inner products need not be preserved. See Section 5.

Definition 3.0.11. Given a state space $\mathbb{H}$ and a QECC $\mathbb{W} \subseteq \mathbb{H}$, a **correctable set of errors** is a quantum operation $\mathcal{E}$ for which there exists a trace preserving quantum operation $\mathcal{R}$ satisfying:

$$\mathcal{R}(\mathcal{E}(\rho)) \propto \rho \quad (48)$$

Definition 3.0.12. A **bit flip error** is an error $(|\varphi\rangle, |\psi\rangle)$ where $|\varphi\rangle$ is an encoding of a message with respect to the encoding $\text{BitFlip}^{\otimes m}$ for some m , such that $X_i |\varphi\rangle = |\psi\rangle$ for some i .

A **phase flip error** is an error $(|\varphi\rangle, |\psi\rangle)$ where $|\varphi\rangle$ is an encoding of a message with respect to the the encoding $\text{PhaseFlip}^{\otimes m}$, such that $Z_i |\varphi\rangle = |\psi\rangle$ for some i .

Let $(|\varphi\rangle, |\psi\rangle)$ be a bit flip error. The following algorithm takes as input $|\psi\rangle$ and reconstructs $|\varphi\rangle$:

Algorithm 3.0.13 (Bit flip correction). ²⁰ Input: a received message $|\psi\rangle$,

1. perform the following projective measurements:

$$\langle\psi| Z_1 Z_2 |\psi\rangle \text{ with resulting state } |\psi'\rangle, \quad (49)$$

followed by

$$\langle\psi'| Z_2 Z_3 |\psi'\rangle \quad (50)$$

let (r_1, r_2) be the pair of results from these measurements.

2. It will be shown that $r_1, r_2 \in \{1, -1\}$, and the resulting state of the second measurement is $|\psi\rangle$.
3. Now retrieve $|\varphi\rangle$ based on the values of r_1, r_2 :
 - if $(r_1, r_2) = (1, 1)$, return $|\psi\rangle$,
 - if $(r_1, r_2) = (-1, 1)$, return $X_1 |\psi\rangle$,
 - if $(r_1, r_2) = (1, -1)$, return $X_3 |\psi\rangle$,
 - if $(r_1, r_2) = (-1, -1)$, return $X_2 |\psi\rangle$

We now prove correctness of Algorithm 3.0.13:

Proof. It will be helpful to first notice:

$$\begin{aligned} Z_1 Z_2 |000\rangle &= |000\rangle & Z_1 Z_2 |001\rangle &= |001\rangle \\ Z_1 Z_2 |010\rangle &= -|010\rangle & Z_1 Z_2 |011\rangle &= -|011\rangle \\ Z_1 Z_2 |100\rangle &= -|100\rangle & Z_1 Z_2 |101\rangle &= -|101\rangle \\ Z_1 Z_2 |110\rangle &= |110\rangle & Z_1 Z_2 |111\rangle &= |111\rangle \end{aligned}$$

Let $|\psi\rangle := a|010\rangle + b|101\rangle$ be a state, ie, an element of $\mathbb{H}^{\otimes 3}$. We perform the measurement $Z_1 Z_2$ followed by $Z_2 Z_3$:

$$\begin{aligned} \langle\psi| Z_1 Z_2 |\psi\rangle &= (a\langle 010| + b\langle 101|) Z_1 Z_2 (a|010\rangle + b|101\rangle) \\ &= (a\langle 010| + b\langle 101|) (-a|010\rangle - b|101\rangle) \\ &= -a^2 - b^2 = -1^{21} \end{aligned}$$

and

$$\begin{aligned} \langle\psi| Z_2 Z_3 |\psi\rangle &= (a\langle 010| + b\langle 101|) Z_2 Z_3 (a|010\rangle + b|101\rangle) \\ &= (a\langle 010| + b\langle 101|) (-a|010\rangle - b|101\rangle) \\ &= -a^2 - b^2 = -1 \end{aligned}$$

²⁰The expression $\langle\psi| Z_1 Z_2 |\psi\rangle$ is the expectation value, not the projective measurement. The projective measurement of $Z_1 Z_2$ uses projectors $P_{\pm} = (I \pm Z_1 Z_2)/2$. Code states are eigenvectors of $Z_1 Z_2, Z_2 Z_3$, so the syndrome is revealed without disturbing the state. Applying $Z_1 Z_2$ as a unitary (as the proof does) is a different operation. See Section 5.

We can infer from the fact that both of these came out as -1 that it was the second bit which was flipped, and so we can correct this. However, what is the impact of this measurement on the state? Again we calculate:

$$\begin{aligned} Z_1 Z_2 (a |010\rangle + b |101\rangle) &= Z_1 (-a |010\rangle + b |101\rangle) \\ &= -a |010\rangle - b |101\rangle \end{aligned}$$

and

$$\begin{aligned} Z_2 Z_3 (-a |010\rangle - b |101\rangle) &= Z_2 (-a |010\rangle + b |101\rangle) \\ &= a |010\rangle + b |101\rangle \end{aligned}$$

and so the measurements (in the end) did not impact our state. $\square$

Later, using the theory of *stabiliser codes*, we will show that in fact single bit flip errors form the full set of correctable errors using $Z_1 Z_2, Z_1 Z_3, Z_2 Z_3$. We can rephrase what we have demonstrated in the language introduced at the beginning of this Section:

Example 3.0.14. ²² Given the state space $\mathbb{H} = (\mathbb{C}^2)^{\otimes 3}$ and QECC $\mathbb{W} \subseteq \mathbb{H}$ given by $\mathbb{W} = \text{im BitFlip}$, a correctable set of errors is $\mathcal{E} = \{Z_1 Z_2, Z_1 Z_3, Z_2 Z_3\}$. The corresponding trace preserving quantum operation $\mathcal{R}$ is given by

$$\mathcal{R}(\rho) = \quad (51)$$

Let $(|\varphi\rangle, |\psi\rangle)$ be a phase flip error. The following algorithm takes as input $|\psi\rangle$ and reconstructs $|\varphi\rangle$:

Algorithm 3.0.15 (Phase flip correction). ²³ Input: a received message $|\psi\rangle$:

1. perform the following projective measurements:

$$\langle \psi | X_1 X_2 | \psi \rangle \text{ with resulting state } |\psi'\rangle \quad (52)$$

followed by

$$\langle \psi' | X_2 X_3 | \psi' \rangle \quad (53)$$

let (r_1, r_2) be the pair of results from these measurements.

2. It will be shown that $r_1, r_2 \in \{1, -1\}$ and the resulting state of the second measurement is $|\psi\rangle$.
3. Now retrieve $|\varphi\rangle$ based on the values of r_1, r_2 :
 - (a) if $(r_1, r_2) = (1, 1)$, return $|\psi\rangle$,
 - (b) if $(r_1, r_2) = (-1, 1)$, return $Z_1 |\psi\rangle$
 - (c) if $(r_1, r_2) = (1, -1)$, return $Z_3 |\psi\rangle$,
 - (d) if $(r_1, r_2) = (-1, -1)$, return $Z_2 |\psi\rangle$

Proof. In fact all our work is already done. We simply note that $Z|+\rangle = |-\rangle, Z|-\rangle = |+\rangle$ (and so phase flip acts like bit flip for $|+\rangle, |-\rangle$), and that in general

$$H^{\otimes n} Z_{i_1} \dots Z_{i_j} H^{\otimes n} = X_{i_1} \dots X_{i_j} \quad (54)$$

The result then follows from the proof of correctness for Algorithm 3.0.13. $\square$

²²The correctable errors are $\{I, X_1, X_2, X_3\}$, not $\{Z_1 Z_2, Z_1 Z_3, Z_2 Z_3\}$. The latter are syndrome observables. See Section 5.

²³Same measurement-versus-application confusion as the bit-flip case: projective measurements use $(I \pm X_1 X_2)/2$ and $(I \pm X_2 X_3)/2$. See Section 5.

What if we wanted to correct an error where we knew the received message corresponded to the intended message by either a bit flip *or* a phase flip. This can be done by combining the two approaches above. Define the following encoding:

Definition 3.0.16. ²⁴ The **Shor encoding** is:

$$\text{Shor} : \mathbb{H} \longrightarrow \mathbb{H}^{\otimes 9} \quad (55)$$

where

$$\text{Shor}(|\psi\rangle) = \text{BitFlip} \circ \text{PhaseFlip} |\psi\rangle \quad (56)$$

Algorithm 3.0.17. ²⁵ On input $|\psi\rangle$:

1. Perform the following projective measurements:

$$\begin{aligned} &\langle\psi| Z_1 Z_2 |\psi\rangle \text{ with resulting state } |\psi'\rangle \\ &\langle\psi'| Z_2 Z_3 |\psi'\rangle \text{ with resulting state } |\psi\rangle \\ &\langle\psi| Z_3 Z_4 |\psi\rangle \text{ with resulting state } |\psi''\rangle \\ &\langle\psi'| Z_4 Z_5 |\psi''\rangle \text{ with resulting state } |\psi\rangle \\ &\langle\psi'| Z_5 Z_6 |\psi\rangle \text{ with resulting state } |\psi'''\rangle \\ &\langle\psi'| Z_6 Z_7 |\psi'''\rangle \text{ with resulting state } |\psi\rangle \\ &\langle\psi'| Z_7 Z_8 |\psi\rangle \text{ with resulting state } |\psi''''\rangle \\ &\langle\psi'| Z_8 Z_9 |\psi''''\rangle \text{ with resulting state } |\psi\rangle \end{aligned}$$

let $(r_1, r_2, r_3, r_4, r_5, r_6, r_7, r_8) \in \mathbb{Z}_2^8$ be the results from these measurements. Notice that there are only three possibilities, all entries are 1, exactly one entry is -1 in which case it is either r_1 or r_8 (with the rest equal to 1) or exactly two values are -1 and the rest are 1 in which case the two -1 entries are neighbours.

2. Then perform the following measurements:

$$\begin{aligned} &\langle\psi| X_1 X_2 X_3 X_4 X_5 X_6 |\psi\rangle \text{ with resulting state } |\psi'\rangle \\ &\langle\psi'| X_4 X_5 X_6 X_7 X_8 X_9 |\psi'\rangle \text{ with resulting state } |\psi\rangle \end{aligned}$$

let $(s_1, s_2) \in \mathbb{Z}_2^2$ be the result of these measurements.

Now retrieve $|\varphi\rangle$ based on the values:

²⁴This composition is not type-correct: $\text{PhaseFlip} : \mathbb{H} \rightarrow \mathbb{H}^{\otimes 3}$ produces three qubits, but $\text{BitFlip} : \mathbb{H} \rightarrow \mathbb{H}^{\otimes 3}$ accepts a single qubit. The correct composition is $\text{Shor} = \text{BitFlip}^{\otimes 3} \circ \text{PhaseFlip}$. See Section 5.

²⁵The listed Z -type measurements are not the Shor stabilisers. Cross-block checks $Z_3 Z_4$ and $Z_6 Z_7$ disturb the encoded information. The correct Z -type generators are $Z_1 Z_2, Z_2 Z_3, Z_4 Z_5, Z_5 Z_6, Z_7 Z_8, Z_8 Z_9$ (within-block only). The recovery table below has length-9 rows where length-8 was claimed, repeats rows, and mixes recovery rules. See Section 5 for the standard formulation.

$(r_1, r_2, r_3, r_4, r_5, r_6, r_7, r_8)$	(s_1, s_2)	Return
$(1, 1, 1, 1, 1, 1, 1, 1)$	$(1, 1)$	$ \psi\rangle$
$(-1, 1, 1, 1, 1, 1, 1, 1)$	$(1, 1)$	$X_1 \psi\rangle$
$(-1, -1, 1, 1, 1, 1, 1, 1)$	$(1, 1)$	$X_2 \psi\rangle$
$\vdots$	$\vdots$	$\vdots$
$(1, 1, 1, 1, 1, 1, -1, -1)$	$(1, 1)$	$X_8 \psi\rangle$
$(1, 1, 1, 1, 1, 1, 1, -1)$	$(1, 1)$	$X_9 \psi\rangle$
$(1, 1, 1, 1, 1, 1, 1, 1)$	$(-1, 1)$	$ \psi\rangle$
$(-1, 1, 1, 1, 1, 1, 1, 1)$	$(-1, 1)$	$Z_1 Z_2 Z_3 X_1 \psi\rangle$
$(-1, -1, 1, 1, 1, 1, 1, 1)$	$(-1, 1)$	$Z_1 Z_2 Z_3 X_2 \psi\rangle$
$\vdots$	$\vdots$	$\vdots$
$(1, 1, 1, 1, 1, 1, -1, -1)$	$(-1, -1)$	$Z_4 Z_5 Z_6 X_8 \psi\rangle$
$(1, 1, 1, 1, 1, 1, 1, -1)$	$(-1, -1)$	$Z_4 Z_5 Z_6 X_9 \psi\rangle$
$(-1, 1, 1, 1, 1, 1, 1, 1)$	$(-1, -1)$	$Z_4 Z_5 Z_6 X_1 \psi\rangle$
$(-1, -1, 1, 1, 1, 1, 1, 1)$	$(-1, -1)$	$Z_4 Z_5 Z_6 X_2 \psi\rangle$
$\vdots$	$\vdots$	$\vdots$
$(1, 1, 1, 1, 1, 1, -1, -1)$	$(-1, -1)$	$Z_4 Z_5 Z_6 X_8 \psi\rangle$
$(1, 1, 1, 1, 1, 1, 1, -1)$	$(-1, -1)$	$Z_4 Z_5 Z_6 X_9 \psi\rangle$
$(1, 1, 1, 1, 1, 1, -1, -1)$	$(-1, -1)$	$Z_4 Z_5 Z_6 X_8 \psi\rangle$
$(1, 1, 1, 1, 1, 1, 1, -1)$	$(-1, -1)$	$Z_4 Z_5 Z_6 X_9 \psi\rangle$
$(-1, 1, 1, 1, 1, 1, 1, 1)$	$(-1, 1)$	$Z_7 Z_8 Z_9 X_1 \psi\rangle$
$(-1, -1, 1, 1, 1, 1, 1, 1)$	$(-1, 1)$	$Z_7 Z_8 Z_9 X_2 \psi\rangle$
$\vdots$	$\vdots$	$\vdots$
$(1, 1, 1, 1, 1, 1, -1, -1)$	$(-1, 1)$	$Z_7 Z_8 Z_9 X_8 \psi\rangle$
$(1, 1, 1, 1, 1, 1, 1, -1)$	$(-1, 1)$	$Z_7 Z_8 Z_9 X_9 \psi\rangle$

Proof. ²⁶ That the Shor algorithm correct bit flip errors is obvious.

Now assume a single phase flip error has occurred, and no bit flip error has occurred. The core observation is the commutativity of the following diagrams for any $|v\rangle \in \{|0\rangle, |1\rangle\}$ (where we think of any “ket” vector $|v\rangle$ as a map $k \rightarrow |v\rangle$) we have written BF for BitFlip:

$$\begin{array}{ccccc}
 k & \xrightarrow{|v\rangle} & \mathbb{H} & \xrightarrow{X} & \mathbb{H} & \xrightarrow{\langle v|} & k \\
 & \searrow |vv\rangle & \downarrow \text{BF} & & \downarrow \text{BF} & \nearrow \langle vv| & \\
 & & \mathbb{H}^{\otimes 3} & \xrightarrow{X \otimes XZ \otimes X} & \mathbb{H}^{\otimes 3} & &
 \end{array} \tag{57}$$

A similar Diagram but with $X \otimes XZ \otimes X$ replaced by $XZ \otimes X \otimes X$ or by $X \otimes X \otimes XZ$ also commutes. Thus, if $|\psi\rangle = Z_i |\varphi\rangle$, defining

$$s(i) = \begin{cases} 1, & i = 1, 2, 3 \\ 2, & i = 4, 5, 6 \\ 3, & i = 7, 8, 9 \end{cases} \tag{58}$$

we have

$$\langle m | \text{BF}^{\otimes 3 \dagger} Z_i^\dagger X_1 X_2 X_3 X_4 X_5 X_6 Z_i \text{BF}^{\otimes 3} | m \rangle = \langle m | Z_{s(i)}^\dagger X_1 X_2 Z_{s(i)} | m \rangle \tag{59}$$

²⁶ “Obvious” is not enough given that the listed measurements are not the standard Shor stabilisers. The combined-error step is correct only up to a global phase, which deserves explicit treatment via Pauli (anti)commutation. See Section 5.

and so we can treat each “block” of three states as a single state, so we know how to interpret the measurements (s_1, s_2) . The last observation to make is commutativity of the following Diagram for all $i = 1, 2, 3$

$$\begin{array}{ccc}
 \mathbb{H}^{\otimes 3} & \xrightarrow{Z_i} & \mathbb{H}^{\otimes 3} \\
 \downarrow \text{BF} & & \downarrow \text{BF} \\
 \mathbb{H}^{\otimes 9} & \xrightarrow{Z_{3i-2}Z_{3i-1}Z_{3i}} & \mathbb{H}^{\otimes 9}
 \end{array} \tag{60}$$

Now say a combination of a bit flip and a phase flip error occurred. That is, say $|\psi\rangle = Z_i X_j |\varphi\rangle$. The error correction will first correct the bit flip which reduces to the previous case. In other words, $X_j^2 |\varphi\rangle = |\varphi\rangle$ is in the image of BitFlip. $\square$

4 The stabiliser formalism

Definition 4.0.1. ²⁷ The **Pauli Group**, denoted G_n , is generated by of all operators $\pm I, iI, X_j, Y_j, Z_j$ for $j = 1, \dots, n$.

Denote by $\mathcal{X}$ denote the following set of operators

$$\mathcal{X} := \{I, X, Y, Z\} \tag{61}$$

I, X, Y, Z . For an arbitrary element $g \in G_n$, let $g^1, \dots, g^n \in \mathcal{X}$ be such that

$$g = \alpha g^1 \otimes \dots \otimes g^n \tag{62}$$

then the sequence $g_i^1, \dots, g_i^n$ is the unique such, and we denote a length $2n$ sequence $x = (x_1, \dots, x_{2n})$ in $\mathbb{Z}_2^{2n}$ by $r(g_i)$ defined by the following schemata:

- $g^j = X$ if and only if $x_j = 1$,
- $g^j = Z$ if and only if $x_{j+n} = 1$,
- $g^j = Y$ if and only if $x_i = x_{j+n} = 1$.

Given a set $\{g_1, \dots, g_k\}$ of elements of the Pauli group, the **check matrix** is the $k \times 2n$ matrix whose i^{th} row is $r(g_i)$. The check matrix is denoted $\text{Check}(g_1, \dots, g_n)$.

Observation 4.0.2. Let (g_1, g_2) be a pair of elements of G_n and let $g_1^1, \dots, g_1^n, g_2^1, \dots, g_2^n \in \mathcal{X}$ be such that

$$g_1 = \alpha_1 g_1^1 \otimes \dots \otimes g_1^n, \quad g_2 = \alpha_2 g_2^1 \otimes \dots \otimes g_2^n \tag{63}$$

we see that g_1 and g_2 commute if and only if the number of times g_1^j and g_2^j are distinct matrices with neither equal to the identity is even.

Defining

$$\Lambda := \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \tag{64}$$

we have the following:

²⁷Cleaner statement: $G_n = \{\alpha P_1 \otimes \dots \otimes P_n : \alpha \in \{\pm 1, \pm i\}, P_j \in \{I, X, Y, Z\}\}$. The body’s binary representation says “ $g^j = X$ iff $x_j = 1$ ”, which fails for $g^j = Y$ (which also has $x_j = 1$); the correct conditions are $x_j = 1 \iff g^j \in \{X, Y\}$ and $z_j = 1 \iff g^j \in \{Y, Z\}$. There is also an index typo x_i for x_j . See Section 5.

Lemma 4.0.3. ²⁸ Let $(g_1, g_2) \in G_n$. Then g_1, g_2 commute if and only if

$$r(g_1)\Lambda r(g_2)^T = 0 \quad (65)$$

Rough sketch. The form of $r(g_1)$:

$$r(g_1) = (X \text{ or } Y \text{ in } g_1 \mid Z \text{ or } Y \text{ in } g_1) \quad (66)$$

and similarly for $r(g_2)$. Thus we have

$$r(g_1)\Lambda r(g_2)^T = (X \text{ or } Y \text{ in } g_1 \mid Z \text{ or } Y \text{ in } g_1) \begin{pmatrix} Z \text{ or } Y \text{ in } g_2 \\ X \text{ or } Y \text{ in } g_2 \end{pmatrix} \quad (67)$$

This is a mathematical incarnation of the requirements specified by Observation 4.0.2. $\square$

The Check matrix is useful for more:

Definition 4.0.4. ²⁹ A set of elements $g_1, \dots, g_r \in G_n$ of the Pauli group G_n are **independent** if the for any j we have, where we write $\hat{g}_j$ for the omission of g_j :

$$\langle g_1, \dots, g_r \rangle \neq \langle g_1, \dots, \hat{g}_j, \dots, g_n \rangle \quad (68)$$

(here, the notation $\langle g_1, \dots, g_n \rangle$ denotes the group generated by these elements).

Lemma 4.0.5. Let $g_1, \dots, g_r \in G_n$ be a set of elements such that $-I \notin \langle g_1, \dots, g_r \rangle$, then the elements $g_1, \dots, g_r$ are independent if and only if $r(g_1), \dots, r(g_r)$ and linearly independent (over the field $\mathbb{Z}_2$).

Proof. Exercise. $\square$

Denote qubit space $\mathbb{C}^2$ by $\mathbb{H}$. Given a subgroup $S \subseteq G_n$ of the Pauli group G_n , we denote by V_S the set of vectors $|\psi\rangle \in \mathbb{H}^{\otimes n}$ such satisfying

$$\forall W \in S, W|\psi\rangle = |\psi\rangle \quad (69)$$

The following Lemma will be used to calculate the dimension of V_S :

Lemma 4.0.6. ³⁰ Let $g_1, \dots, g_k$ be independent elements of the Pauli group G_n and denote by S the group they generate. Assume $-I \notin S$. Then for each $i = 1, \dots, k$ there exists $g \in G_n$ such that g anti-commutes with g_i and commutes with all g_j such that $i \neq j$.

Proof. The set $r(g_1), \dots, r(g_k)$ is linearly independent by Lemma 4.0.5, thus the check matrix of $g_1, \dots, g_k$ has k linearly independent columns. So, there exists a vector $x \in \mathbb{Z}_2^k$ such that

$$\text{Check}(g_1, \dots, g_n)\Lambda x = e_i \quad (70)$$

where e_i is the i^{th} standard basis vector of $\mathbb{Z}_2^k$. Let g be such that $r(g)^T = x$. The result follows from Lemma 4.0.5. $\square$

Theorem 4.0.7. ³¹ Let $S = \langle g_1, \dots, g_k \rangle$ and say $-I \notin S$. Then $\dim V_S = 2^{n-k}$.

²⁸The product is computed over $\mathbb{F}_2$ (modulo 2). See Section 5.

²⁹Index typo: the right-hand side should have g_r , not g_n . See Section 5.

³⁰The proof says “ k linearly independent columns”; should read “rows”. The vector x should lie in $\mathbb{F}_2^{2^n}$, not $\mathbb{F}_2^k$ (the wrong dimension). See Section 5.

³¹The conclusion is correct, but the proof has issues: “ I is a projector onto an n -dimensional space” should read “ 2^n -dimensional” (since n qubits give $(\mathbb{C}^2)^{\otimes n}$ of dimension 2^n). Independence requires g_j Hermitian with eigenvalues ± 1 , which uses $-I \notin S$ to rule out phase- i generators. See Section 5.

Proof. We notice that $(1/2)(I + g_j)$ is the projector onto the $+1$ -Eigenspace of g_j . We let $x = (x_1, \dots, x_k) \in \mathbb{Z}_2^k$ and define the operator

$$P_S^x := 1/2^k \prod_{j=1}^k (I + (-1)^{x_j} g_j) \quad (71)$$

By Lemma 4.0.6 we have for each g_j there exists g_{x_j} such that $g_{x_j} g_j g_{x_j}^{-1} = -g_j$. Let $g_x = g_{x_1} \dots g_{x_k}$, then

$$\begin{aligned} g_x P_S^{(0, \dots, 0)} g_x^{-1} &= 1/2^k \prod_{j=1}^k (g_{x_j} g_j g_{x_j}^{-1} + g_{x_j} g_j g_{x_j}^{-1}) \\ &= P_S^x \end{aligned}$$

Thus there is an isomorphism

$$\text{im } P_S^x \cong \text{im } P_S^{(0, \dots, 0)} \quad (72)$$

Since $\text{im } P_S \cong V_S$ we have $\dim \text{im } P_S^x = \dim V_S$. Finally we note that

$$I = \sum_{x \in \mathbb{Z}_2^k} P_S^x \quad (73)$$

The operator I is a projector onto an n -dimensional space, and $\sum_{x \in \mathbb{Z}_2^k} P_S^x$ is a sum of 2^k orthogonal projectors all of the same dimension as V_S , thus the only possibility is $\dim V_S = 2^{n-k}$. $\square$

5 AI-generated errata

This section was produced by a large language model. It corrects statements and proof steps in the body. Each subsection mirrors a section above; corrected statements supersede the originals.

5.1 Mathematical definitions

Qubit

The standard convention is: $\mathbb{C}^2$ is a single-qubit state space, and $(\mathbb{C}^2)^{\otimes n}$ is the n -qubit state space. The opening definition should read “an n -qubit state space is $(\mathbb{C}^2)^{\otimes n}$ ”.

Trace

For the formula $\text{Tr}(A) = \sum_i \langle e_i | A | e_i \rangle$ to hold, the basis $\{|e_i\rangle\}$ must be *orthonormal*, not merely orthogonal. For an orthogonal basis $\{|v_i\rangle\}$ the formula carries norm factors. Throughout the document, finite-dimensionality of $\mathbb{H}$ should be assumed at the start, since the trace is otherwise only defined for trace-class operators.

Measurements

The completeness condition is $\sum_{m \in \mathcal{M}} M_m^\dagger M_m = I$, with subscripts on each M_m . The post-measurement state $M_m |\psi\rangle / \sqrt{p(m)}$ is defined only when $p(m) > 0$.

Projectors

A projector in quantum mechanics is an *orthogonal* projection: $P^2 = P$ and $P^\dagger = P$. The bare condition $P^2 = P$ defines an algebraic idempotent, which is not what projective measurements use. The projectors P_m from the spectral theorem of a Hermitian operator are automatically orthogonal.

Extension to a unitary

The lemma stating that $U : W \rightarrow V$ extends to a unitary is incorrectly stated. If U is literally unitary onto V then $\dim W = \dim V$, forcing $W = V$. The intended hypothesis is that U is an *isometry* $W \rightarrow V$.

Lemma 5.1.1. *Fix a finite-dimensional Hilbert space V , a subspace $W \subseteq V$, and an isometry $U : W \rightarrow V$. Then U extends to a unitary $U' : V \rightarrow V$.*

Proof. $U(W) \subseteq V$ has dimension $\dim W$. Choose any unitary $W^\perp \rightarrow U(W)^\perp$, and define U' to equal U on W and the chosen unitary on $W^\perp$. Then U' is unitary on V . $\square$

The original proof $U' = \text{Proj}_W \circ U$ is wrong: domain and codomain do not match, and the result is not a unitary extension.

Naimark dilation

The proposition is essentially correct but the proof has many notation and tensor-order errors. A clean version:

Proposition 5.1.2. *Fix a measurement $\{M_m\}_{m \in \mathcal{M}}$ on $\mathbb{H}$. Let Q have orthonormal basis $\{|m\rangle : m \in \mathcal{M}\}$ and define $V : \mathbb{H} \rightarrow \mathbb{H} \otimes Q$ by*

$$V |\psi\rangle = \sum_m M_m |\psi\rangle \otimes |m\rangle.$$

Then $V^\dagger V = \sum_m M_m^\dagger M_m = I$, so V is an isometry. Fix $|\varphi\rangle \in Q$ and extend

$$U(|\psi\rangle \otimes |\varphi\rangle) = \sum_m M_m |\psi\rangle \otimes |m\rangle$$

to a unitary U on $\mathbb{H} \otimes Q$ (using the corrected isometry-extension lemma). Measuring the ancilla with projectors $P_n = I_{\mathbb{H}} \otimes |n\rangle\langle n|$ on $U(|\psi\rangle \otimes |\varphi\rangle)$ gives

$$p(n) = \langle \psi | M_n^\dagger M_n | \psi \rangle.$$

The original proof switches between $Q \otimes \mathbb{H}$ and $\mathbb{H} \otimes Q$, writes $I_q \otimes |m\rangle\langle m|$ where $I_{\mathbb{H}} \otimes |m\rangle\langle m|$ is intended, omits subscripts on $M_m^\dagger M_{m'}$, and has a wrong final ket in the inner-product calculation.

5.2 Density operator

Pure versus mixed

Purity is a property of the density operator, not the ensemble. The density operator ρ is *pure* iff $\rho = |\psi\rangle\langle\psi|$ for some unit vector, equivalently $\text{Tr}(\rho^2) = 1$. An ensemble of length > 1 can still give a pure ρ .

Measurement update (operator order)

The post-measurement density operator is

$$\rho_m = \frac{M_m \rho M_m^\dagger}{\text{Tr}(M_m^\dagger M_m \rho)},$$

not $M_m^\dagger \rho M_m$. With the correct ordering, the pure-state and density-operator measurement rules agree:

$$\frac{M_m |\psi\rangle\langle\psi| M_m^\dagger}{p(m)} = \left(\frac{M_m |\psi\rangle}{\sqrt{p(m)}} \right) \left(\frac{M_m |\psi\rangle}{\sqrt{p(m)}} \right)^\dagger.$$

The displayed expression $\rho_m = \sum_{i=1}^n \frac{M_m \rho M_m^\dagger}{\text{Tr}(M_m^\dagger M_m \rho)}$ in the derivation also has an erroneous factor of n ; the sum disappears once ρ is substituted.

Time evolution

The first evolved state in the sequence is $U_1 \rho U_1^\dagger$, not $U \rho U^\dagger$:

$$(\rho, U_1 \rho U_1^\dagger, U_2 U_1 \rho U_1^\dagger U_2^\dagger, \dots, U_n \dots U_1 \rho U_1^\dagger \dots U_n^\dagger).$$

Equivalence of ensembles

The proposition as written is mathematically incoherent: the notation $\{p_i |i\rangle\langle i|\}$ is not standard ensemble notation, the indices and bras are conflated, and the proof omits bras in spectral decompositions. The correct theorem is the standard unitary-freedom-of-ensembles theorem.

Proposition 5.2.1. *Fix two ensembles $\{p_i, |\psi_i\rangle\}_{i=1}^m$ and $\{q_j, |\varphi_j\rangle\}_{j=1}^r$. Define unnormalised vectors*

$$|\tilde{\psi}_i\rangle = \sqrt{p_i} |\psi_i\rangle, \quad |\tilde{\varphi}_j\rangle = \sqrt{q_j} |\varphi_j\rangle,$$

and pad the shorter list with zero vectors so both have the same length N . The two ensembles determine the same density operator iff there exists a unitary $U = (u_{ij})$ with

$$|\tilde{\psi}_i\rangle = \sum_j u_{ij} |\tilde{\varphi}_j\rangle.$$

For ensembles of unequal length without padding, the appropriate object is an isometry or partial isometry rather than a square unitary.

The preceding observation also has errors. The diagonal form is

$$\sum_i p_i |\psi_i\rangle \langle \psi_i| = \sum_j \lambda_j |\varphi_j\rangle \langle \varphi_j|,$$

not $\sum_j \lambda_j |\varphi_j\rangle$ (a missing bra). Coefficient comparison gives

$$\lambda_j \delta_{jk} = \sum_i p_i \alpha_{ij} \overline{\alpha_{ik}},$$

not $\sum_i p_i \alpha_{ij} \alpha_{ij}^\dagger$.

5.3 Partial trace

The intermediate calculation in the Bell-state example duplicates the $|11\rangle \langle 00|$ line. The correct partial traces are:

$$\text{Tr}_2(|11\rangle \langle 00|) = \text{Tr}(F_{01})E_{01} = 0, \quad \text{Tr}_2(|00\rangle \langle 11|) = \text{Tr}(F_{10})E_{10} = 0, \quad \text{Tr}_2(|11\rangle \langle 11|) = \text{Tr}(F_{11})E_{11} = |1\rangle \langle 1|.$$

The final result $\text{Tr}_2(\rho) = I/2$ is correct.

Adjunction

The standard finite-dimensional tensor-Hom adjunction is

$$\text{Hom}(V \otimes U, W) \cong \text{Hom}(U, V^* \otimes W),$$

not $\text{Hom}(V, W \otimes V^*)$. The right adjoint of $V \otimes -$ is $V^* \otimes -$.

5.4 Quantum operations

A quantum operation is a linear map $\mathcal{E} : L(\mathbb{H}_p) \rightarrow L(\mathbb{H}_p)$ on operators, not $\mathbb{H}_p \rightarrow \mathbb{H}_p$. Trace-preserving completely positive such maps are called quantum channels.

Kraus form

For a pure environment state $\rho_e = |e_0\rangle \langle e_0|$, the Kraus operators are

$$E_i = (I \otimes \langle i|)U(I \otimes |e_0\rangle),$$

giving $\mathcal{E}(\rho) = \sum_i E_i \rho E_i^\dagger$. The original derivation conflates the summation index with the fixed environment state, writes $\rho \otimes |i\rangle \langle i|$ where $\rho \otimes |e_0\rangle \langle e_0|$ is intended, and uses an unnecessary multiplication map.

Lemma 5.4.1. *A trace-preserving quantum operation $\mathcal{E}$ admits an operator-sum representation $\mathcal{E}(\rho) = \sum_i E_i \rho E_i^\dagger$ with $\sum_i E_i^\dagger E_i = I$. The Kraus operators $\{E_i\}$ are not unique.*

For mixed environment $\rho_e = \sum_a p_a |e_a\rangle \langle e_a|$, the Kraus operators are $E_{ia} = \sqrt{p_a}(I \otimes \langle i|)U(I \otimes |e_a\rangle)$.

5.5 Quantum error correction

Basis convention

The standard convention (consistent with $Z = \text{diag}(1, -1)$) is $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$, giving $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$. The body's reversed convention contradicts later calculations and should be corrected.

Bell states versus Hadamard-basis states

The states $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ are the Hadamard-basis (or X -basis) states, not Bell states. Bell states are two-qubit entangled states such as $(|00\rangle + |11\rangle)/\sqrt{2}$.

Encoding

A quantum encoding $\iota : \mathbb{H} \rightarrow \mathbb{H}^{\otimes n}$ should be a linear isometry, not merely an injective map, in order to preserve superpositions and inner products.

Bit-flip correction: measurement versus application

The proof confuses two distinct operations. Writing $\langle\psi| Z_1 Z_2 |\psi\rangle$ gives the expectation value of the observable, not the projective measurement. The projective measurement of $Z_1 Z_2$ uses projectors

$$P_{\pm} = \frac{I \pm Z_1 Z_2}{2},$$

and the post-measurement state is $P_{\pm} |\psi\rangle / \sqrt{p(\pm)}$. Code states are eigenvectors of $Z_1 Z_2, Z_2 Z_3$, so the syndrome measurement reveals the syndrome without disturbing the state. Applying $Z_1 Z_2$ as a unitary is a different operation entirely.

The amplitude error $-a^2 - b^2 = -1$ should read $-|a|^2 - |b|^2 = -1$ for general $a, b \in \mathbb{C}$. The bra should be $\bar{a} \langle 010| + \bar{b} \langle 101|$.

Correctable set of errors

For the bit-flip code, the correctable errors are $\{I, X_1, X_2, X_3\}$, not $\{Z_1 Z_2, Z_1 Z_3, Z_2 Z_3\}$. The latter are syndrome observables. A correct recovery uses syndrome projectors $P_{\pm\pm}$ and corrections I, X_1, X_3, X_2 for syndromes $(++, -, +-, --)$ respectively.

Phase-flip

The same measurement-versus-application confusion appears. The projective measurements use $(I \pm X_1 X_2)/2$ and $(I \pm X_2 X_3)/2$. The identity $H^{\otimes n} Z_i H^{\otimes n} = X_i$ is correct.

5.6 Shor code

Encoding type

The composition $\text{BitFlip} \circ \text{PhaseFlip}$ is not type-correct: $\text{PhaseFlip} : \mathbb{H} \rightarrow \mathbb{H}^{\otimes 3}$ produces three qubits, but $\text{BitFlip} : \mathbb{H} \rightarrow \mathbb{H}^{\otimes 3}$ accepts a single qubit. The correct composition is

$$\text{Shor} = \text{BitFlip}^{\otimes 3} \circ \text{PhaseFlip}.$$

Explicitly,

$$|0\rangle \mapsto \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)^{\otimes 3}, \quad |1\rangle \mapsto \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)^{\otimes 3}.$$

Stabiliser measurements

The Shor code does not use $Z_i Z_{i+1}$ for $i = 1, \dots, 8$. The standard Z -type stabiliser generators are

$$Z_1 Z_2, Z_2 Z_3, Z_4 Z_5, Z_5 Z_6, Z_7 Z_8, Z_8 Z_9,$$

checking bit flips within each block of three. Cross-block checks $Z_3 Z_4, Z_6 Z_7$ are not stabilisers and disturb the encoded information.

The standard X -type stabiliser generators are

$$X_1X_2X_3X_4X_5X_6, X_4X_5X_6X_7X_8X_9.$$

This gives 8 stabiliser generators total, not 10.

Recovery table

The original recovery table has length-9 rows where length-8 was promised, repeats rows, has impossible syndrome patterns, and mixes bit-flip and phase-flip recovery inconsistently. The correct rule:

- Use the six Z -type checks to identify the bit-flip location and apply X_i .
- Use the two X -type checks to identify the block with a phase flip; apply Z to any qubit in that block (or equivalently $Z_iZ_{i+1}Z_{i+2}$ to the whole block).

The phase-syndrome interpretation:

(s_1, s_2)	phase-error block
$(+1, +1)$	none
$(-1, +1)$	block 1 (qubits 1,2,3)
$(-1, -1)$	block 2 (qubits 4,5,6)
$(+1, -1)$	block 3 (qubits 7,8,9)

Combined errors

For a combined error $Z_iX_j|\varphi\rangle$, the correction is correct up to a global phase. This requires using that Pauli operators either commute or anticommute, and anticommutation contributes only an unobservable global phase after both corrections.

5.7 Stabiliser formalism

Pauli group

The cleanest statement:

$$G_n = \{\alpha P_1 \otimes \cdots \otimes P_n : \alpha \in \{\pm 1, \pm i\}, P_j \in \{I, X, Y, Z\}\}.$$

Binary representation

For $g = \alpha g^1 \otimes \cdots \otimes g^n$, the binary vector $r(g) = (x_1, \dots, x_n, z_1, \dots, z_n)$ satisfies

$$x_j = 1 \iff g^j \in \{X, Y\}, \quad z_j = 1 \iff g^j \in \{Y, Z\}.$$

The original “ $g^j = X$ iff $x_j = 1$ ” fails for $g^j = Y$, since Y also has $x_j = 1$. The single-qubit map is $I \mapsto (0, 0)$, $X \mapsto (1, 0)$, $Z \mapsto (0, 1)$, $Y \mapsto (1, 1)$.

There is also an index typo: “ $x_i = x_{j+n} = 1$ ” should read “ $x_j = x_{j+n} = 1$ ”.

Symplectic commutation

The condition $r(g_1)\Lambda r(g_2)^T = 0$ should be computed over $\mathbb{F}_2$ (i.e. modulo 2), with

$$\Lambda = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}.$$

Independence

The notation in the independence definition has a typo: $\langle g_1, \dots, \widehat{g}_j, \dots, g_n \rangle$ should be $\langle g_1, \dots, \widehat{g}_j, \dots, g_r \rangle$. The hypothesis $-I \notin \langle g_1, \dots, g_r \rangle$ is essential for binary-vector independence to coincide with Pauli-group independence.

Anticommuting Pauli

The check matrix H is $k \times 2n$. Linear independence of the $r(g_i)$ means H has k linearly independent *rows* (not columns), so $\text{rank } H = k$. Since Λ is invertible, $H\Lambda$ has rank k , so $x \mapsto H\Lambda x$ from $\mathbb{F}_2^{2n}$ to $\mathbb{F}_2^k$ is surjective. For each $e_i \in \mathbb{F}_2^k$ there exists $x \in \mathbb{F}_2^{2n}$ with $H\Lambda x = e_i$. The original “ $x \in \mathbb{Z}_2^k$ ” has the wrong dimension.

Stabiliser dimension theorem

The conclusion $\dim V_S = 2^{n-k}$ is correct. The proof needs repair.

Each g_j is Hermitian with eigenvalues ± 1 (since $-I \notin S$, no element has order-4 phase). The simultaneous eigenspace projectors are

$$P_x = \frac{1}{2^k} \prod_{j=1}^k (I + (-1)^{x_j} g_j).$$

For each j , choose $h_j \in G_n$ anticommuting with g_j and commuting with g_ℓ for $\ell \neq j$ (Lemma 4.0.6); for $x = (x_1, \dots, x_k)$ set $h_x = \prod_{j: x_j=1} h_j$. Then conjugation by h_x sends P_0 to P_x , so all P_x have equal rank.

The physical Hilbert space is $(\mathbb{C}^2)^{\otimes n}$, dimension 2^n (the original “the operator I is a projector onto an n -dimensional space” is wrong: n qubits give a 2^n -dimensional space). Since $I = \sum_x P_x$ is a sum of 2^k orthogonal projectors of equal rank summing to 2^n ,

$$\dim V_S = \text{rank } P_0 = \frac{2^n}{2^k} = 2^{n-k}.$$

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